Gamma function
Beta function.

① $\Gamma(x) = \int_0^\infty e^{-t}\, t^{x-1}\, dt$ $\quad x > 0$

$\lim_{x \to \pi}$

ⓘ $\Gamma(x+1) = x\Gamma(x)$ $\quad \Gamma(3) = 2\Gamma(2)$

x is any real number, $\neq 0, -1, -2, \ldots$

$\Gamma\left(\frac{1}{3}\right), \Gamma\left(\frac{1}{4}\right), \Gamma\left(-\frac{1}{2}\right), \Gamma\left(-\frac{14}{4}\right) - \ldots$

$\Gamma(0)$ ✗ $\quad \Gamma(-1)$ ✗

$\Gamma(x)$ is generalized factorial.

ⓘⓘ $\Gamma(n+1) = n!$

ⓘⓘⓘ $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$\Gamma(x+1) = x\Gamma(x)$

ⓘⓥ $\Gamma(x) = \dfrac{\Gamma(x+n)}{x(x+1)(x+2)\cdots(x+n-1)}$

$n - 1 + 1 = n$ terms.

n term?

ⓥ $\Gamma(x)\Gamma(1-x) = \dfrac{\pi}{\sin(\pi x)}$ $\quad$ x is non-integer

- Prove Identities.
- Evaluating integrals.

Prove relations/identities.

(a) $\Gamma\left(n+\frac{1}{2}\right) = \frac{(2n)!}{2^{2n}(n)!}\sqrt{\pi}$

$n = 1, 2, \ldots$

$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1}\, dt$

$\Gamma(x+1) = x\Gamma(x)$

$\Gamma(n+1) = n!$

$\Gamma\left(n+\frac{1}{2}\right) = \left(n-\frac{1}{2}\right)\Gamma\left(n-\frac{1}{2}\right)$

$= \left(n-\frac{1}{2}\right)\left(n-\frac{3}{2}\right)\Gamma\left(n-\frac{3}{2}\right)$

$= \left(n-\frac{1}{2}\right)\left(n-\frac{3}{2}\right)\cdots\cdots\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)\sqrt{\pi}$

$= \left(\frac{2n-1}{2}\right)\left(\frac{2n-3}{2}\right)\cdots\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\sqrt{\pi}$ ✓

$= \frac{(2n-1)(2n-3)\cdots(3)(1)\sqrt{\pi}}{2^n}$

$= \frac{(2n)(2n-1)(2n-2)(2n-3)\cdots(4)(3)(2)(1)\sqrt{\pi}}{2^n (2n)(2n-2)(2n-4)\cdots(4)(2)}$

$= \frac{(2n)!\sqrt{\pi}}{2^n 2^n (n)(n-1)(n-2)\cdots(1)}$

$= \frac{(2n)!\sqrt{\pi}}{2^{2n} n!}$ (✳)

$(2n)!$

$n=1$: $\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right)$

$n=2$: $\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$

$n=3$: $\Gamma\left(\frac{7}{2}\right) = \frac{5}{2}\frac{3}{2}\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$

$n=1$ $\quad 2(1) = 2^1$

$n=2$ $\quad = (4)(2)$

$n=3$ $\quad = (6)(4)(2)$

→ $\Gamma(-x) = \frac{\Gamma(-1+x)}{-x}$ ←

$\Gamma(x+1) = x\Gamma(x)$ $\quad x \to -x$

$\Gamma(-x+1) = -x\Gamma(-x)$

(b) $\Gamma\left(\frac{1}{2}-n\right) = \frac{(-1)^n 2^{2n-1}(n-1)!\sqrt{\pi}}{(2n-1)!}\Gamma\left(\frac{1}{2}\right)$ $\quad n=1, 2, \cdots$

Let $x = \frac{1}{2}-n$ $\quad \Gamma(x) = \frac{\Gamma(x+n)}{x(x+1)\cdots(x+n-1)}$

$$\Gamma\left(\tfrac{1}{2}-n\right) = \frac{\Gamma\left(\tfrac{1}{2}-n+n\right)}{\left(\tfrac{1}{2}-n\right)\left(\tfrac{1}{2}-n+1\right)\cdots\left(\tfrac{1}{2}-n+n-1\right)}$$

$$\Gamma\left(\tfrac{1}{2}-n\right) = \frac{\Gamma\left(\frac{1}{2}\right)}{\left(\frac{1}{2}-n\right)\left(\frac{3}{2}-n\right)\left(\frac{5}{2}-n\right)\cdots\left(\frac{-3}{2}\right)\left(\frac{-1}{2}\right)}$$ ← n-terms.

$$= \frac{\sqrt{\pi}}{(-1)^n\left(n-\frac{1}{2}\right)\left(n-\frac{3}{2}\right)\left(n-\frac{5}{2}\right)\cdots\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)} \qquad \frac{1}{(-1)^n} = (-1)^n$$

$$= \frac{(-1)^n\sqrt{\pi}}{\left(\frac{1}{2}\right)^n(2n-1)(2n-3)\cdots(3)(1)} = \frac{2^n(-1)^n\sqrt{\pi}}{(2n-1)(2n-3)\cdots(3)(1)}$$

$$= \frac{(-1)^n 2^n (2n-2)(2n-4)\cdots(4)(2)\sqrt{\pi}}{(2n-1)(2n-2)(2n-3)(2n-4)\cdots(4)(3)(2)(1)}$$

$$= (-1)^n 2^n \frac{2^{n-1}(n-1)!\sqrt{\pi}}{(2n-1)!}$$

$\Gamma\left(\frac{3}{2}-n\right) = ?$ $\quad \frac{3}{2}-n$ $\quad 1+\left(\frac{1}{2}-n\right)$ $\quad x = n-\frac{1}{2}$

$$\Gamma(-x) = \frac{\Gamma(1-x)}{-x}$$

$$\Gamma\left(\tfrac{1}{2}-n\right) = \frac{\Gamma\left(\frac{3}{2}-n\right)}{\frac{1}{2}-n}$$

$\Gamma\left(-\frac{7}{2}\right) =$

(d) $\frac{\Gamma(2x)}{\Gamma(x)} = \frac{2^{2x-1}\Gamma\left(x+\frac{1}{2}\right)}{\sqrt{\pi}}$ → This is also known as **Legendre's Duplication Formula**

relations · integration.

(j) $\Gamma(x) = \Gamma(x+1) - x\Gamma'(x)$ ← $\Gamma(x) = \int_0^\infty \ln t \cdots$

$\Gamma(x+1) = x\Gamma(x)$

$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt$

$\frac{d}{dx} a^x = a^x \ln a$

$\frac{d\Gamma(x)}{dx} = \int_0^\infty e^{-t} t^{x-1} \ln t \, dt = \Gamma'$ → $\Gamma(x)$, $\Gamma\left(\frac{1}{2}\right)$

$\Gamma(x+1) = x\Gamma(x)$

$\Gamma'(x+1) = \Gamma(x) + x\Gamma'(x)$ ⇒ $\Gamma'(x) = \Gamma'(x+1) - y\Gamma'(x)$

$\Gamma(x)$

$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt$

$\beta = \int_0^1$

$\beta = \int_0^\infty$

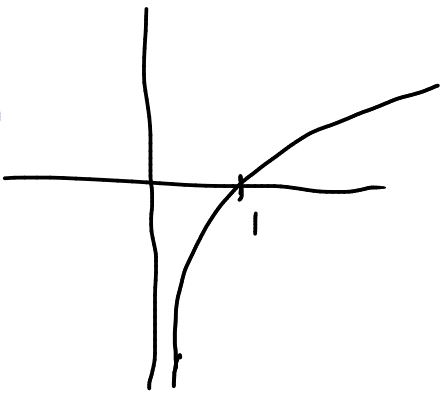
$\beta = \int_0^{\frac{u}{2}}$

Evaluate

(f) $\int_0^1 \left(\ln \frac{1}{u}\right)^{x-1} du.$

Let $\ln u^{-1} = y$ ⇒ $-\ln u = y$ ← $u = e^{-y}$

$-\frac{1}{u} du = dy$



when $u = 0$ ⇒ $y = \infty$

when $u = 1$ ⇒ $y = 0$

$$\int_0^1 \left(\ln \tfrac{1}{u}\right)^{x-1} du = \int_\infty^0 y^{x-1}(-u\,dy)$$

$$= \int_0^\infty e^{-y} y^{x-1}\, dy = \boxed{\Gamma(x)}$$

(a) $\int_0^\infty e^{2ax-x^2}dx = \frac{1}{2}\sqrt{\pi}e^{a^2}$

e^{2ax} e^{-x^2}

Let $u = 2ax - x^2$

$du = (2a - 2x)dx$

$dx = \dfrac{du}{2(a-x)}$

$x^2 - 2ax + a^2 = (x-a)^2$ completing the square

$2ax - x^2 + a^2 - a^2$

$-(x^2 - 2ax + a^2) + a^2$

$$\int_0^\infty e^{-(x-a)^2} e^{a^2} dx$$

$$e^{a^2}\int_0^\infty e^{-(x-a)^2} dx$$

Let $u = (x-a)^2$

$du = 2(x-a)\,dx$

$du = 2\sqrt{u}\,dx$

$x-1$

u

$$e^{a^2}\int_0^\infty e^{-u}\,\frac{du}{2\sqrt{u}} = \frac{1}{2}e^{a^2}\int_0^\infty u^{-\frac{1}{2}}e^{-u}\,du$$

$$\frac{1}{2}e^{a^2}\Gamma\left(-\tfrac{1}{2}+1\right) = \frac{\sqrt{\pi}e^{a^2}}{2}$$

(b) $\displaystyle\int_0^\infty e^{-2x}x^6\,dx = \frac{45}{8}$

Let $u = 2x$ $\quad x=0 \Rightarrow u=0$, $\quad x=\infty \Rightarrow u=\infty$

$du = 2\,dx$

$$\int_0^\infty e^{-u}\left(\frac{u}{2}\right)^6\frac{du}{2}$$

$$= \frac{1}{2^7}\int_0^\infty e^{-u}u^6\,du$$

$\Gamma(7) = 6!$

$$\frac{1}{2^7}\Gamma(7) = \frac{6!}{2^7} \to \frac{45}{8}$$

(e) $\Gamma(x) = (\ln b)^x \int_0^\infty t^{x-1} b^{-t}\, dt, \quad x > 0,\ b > 1$

$$\Gamma(x) = \int_0^\infty e^{-u} u^{x-1}\, du$$

Let $e^{-u} = b^{-t}$

$\frac{1}{e^u} = \frac{1}{b^t} \Rightarrow e^u = b^t$

$u = t \ln b \quad \Leftrightarrow \quad t = \frac{u}{\ln b}$

$du = (\ln b)\, dt$

$$\Gamma(x) = \int_0^\infty e^{-u} u^{x-1}\, du$$

when $u = 0, \ t = 0$

$u = \infty, \ t = \infty$

$$\Gamma(x) = \int_0^\infty b^{-t} (t \ln b)^{x-1} (\ln b)\, dt$$

$$(\ln b)^x \int_0^\infty b^{-t} t^{x-1}\, dt = \Gamma(x)$$

Beta- function

$$\beta(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$$

$$\beta(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

(l) $\beta(x,x) = 2^{1-2x}\beta\left(x,\frac{1}{2}\right)$,

$$\frac{\Gamma(x)\Gamma(x)}{\Gamma(2x)} = 2^{1-2x}\frac{\Gamma(x)\Gamma(\frac{1}{2})}{\Gamma(x+\frac{1}{2})}$$

Duplication formula

$$\int_0^1 , \quad \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

(n) $\beta(n,n)\beta\left(n+\frac{1}{2},n+\frac{1}{2}\right) = \pi 2^{1-4n}n^{-1}, \quad n = 0,1,2,\ldots$

$$\frac{\Gamma(n)\Gamma(n)}{\Gamma(2n)} \quad \frac{\Gamma(n+\frac{1}{2})\Gamma(n+\frac{1}{2})}{\Gamma(2n+1)}$$

$$\Gamma\left(n+\tfrac{1}{2}\right)\cdots = \frac{(2n)!\sqrt{\pi}}{2^{2n}(n!)}$$

$$\frac{(n-1)!\,(n-1)!}{(2n-1)!}\,\frac{(2n)!\,\pi}{(2n)!}\,\frac{1}{2^{4n}(n!)^2}$$

$-n! = n(n-1)!$

$-(2n)! = 2n(2n-1)!$

$$\frac{(n-1)!\,(n-1)!}{(2n-1)!}\,\frac{(2n)!\,\pi}{2^{4n}\,n!\,n!}$$

$$\frac{(n-1)!\,(n-1)!}{(2n-1)!}\,\frac{2n(2n-1)!\,\pi}{2^{4n}\,n(n-1)!\,n(n-1)!} = \frac{2\pi}{4n}\,\frac{1}{2^{n}\,n}$$

$$= 2^{1-4n}\,n^{-1}\,\pi$$

- Evaluating integrals.

- $\beta(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$

(j) $\int_0^a x^4\sqrt{a^2-x^2}\,dx$ $=$ $\int_0^a x^4\left(a^2-x^2\right)^{\frac{1}{2}}\,dx$

Let $a=1$

$$a\int_0^a x^4\left(1-\left(\frac{x}{a}\right)^2\right)^{\frac{1}{2}}dx$$

$$a\,a^4\int_0^a \left[\left(\frac{x}{a}\right)^2\right]^2\left(1-\left(\frac{x}{a}\right)^2\right)^{\frac{1}{2}}dx$$

$\beta\left(3,\frac{3}{2}\right)$

Let $u=\left(\frac{x}{a}\right)^2 \Rightarrow du = \frac{2x}{a^2}\,dx$

$\sqrt{ua^2} = x = a\sqrt{u}$, $\therefore$ $dx = \frac{a}{2\sqrt{u}}\,du$

when $x=0$, $u=0$

$x=a$, $u=1$

$$\beta = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$$

$x-1=\frac{3}{2}$

$x=\frac{5}{2}$

$\frac{1}{2}=y-1$, $y=\frac{3}{2}$

$$a^5\int_0^1 u^2(1-u)^{\frac{1}{2}}\frac{a}{2\sqrt{u}}\,du$$

$$= \frac{a^6}{2}\int_0^1 u^{\frac{3}{2}}(1-u)^{\frac{1}{2}}\,du = \frac{a^6}{2}\beta\left(\frac{5}{2},\frac{3}{2}\right)$$

$$= \frac{a^6}{2}\,\frac{\Gamma(5/2)\Gamma(3/2)}{\Gamma(4)}$$

$$\frac{a^6}{2}\,\frac{\Gamma(5/2)\,\Gamma(3/2)}{\Gamma(4)}$$

$$= \frac{a^6}{2}\,\frac{\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\sqrt{\pi}\left(\frac{1}{2}\right)\sqrt{\pi}}{3\times 2}$$

$$\boxed{a^6\,\frac{3\pi}{3\times 2\times 4}}$$

(#)

(k) $$\int_0^\infty e^{-st}\left(1-e^{-t}\right)^n dt = \frac{n!\,\Gamma(s)}{\Gamma(s+n+1)},$$

$$\Gamma(x) = \int_0^\infty e^{-t}\,t^{x-1}\,dt$$

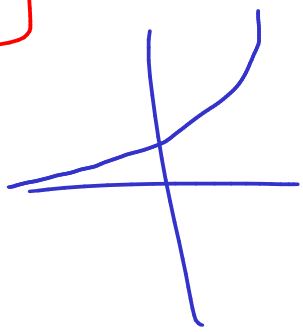
$$\beta(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$$

$$\int_0^\infty e^{-st}\left(1-e^{-t}\right)^n dt$$

Let $u = e^{-t} \Rightarrow du = -u\,dt$

when $t=0 \Rightarrow u=1$

$t=\infty \Rightarrow u=0$



$$\int_1^0 u^s\,(1-u)^n\,\frac{(du)}{-u}$$

$$= \int_0^1 u^{s-1}(1-u)^n\,du = \beta(s,n+1)$$

$$= \frac{\Gamma(s)\,\Gamma(n+1)}{\Gamma(n+s+1)} = \frac{n!\,\Gamma(s)}{\Gamma(n+s+1)}$$

$$\beta(x,y) = \int_0^\infty \frac{u^{x-1}}{(1+u)^{x+y}}\,du \quad \leftarrow \quad x-1 = p-1$$

$$p+y = 1$$

$$y = p-1$$

(f) $$\int_0^\infty \frac{x^{p-1}}{(1+x)^1}\,dx = \pi \csc p\pi, = \beta(p, p-1)$$

$$\int_0^\infty \frac{t^{p-1}}{(1+t)^1}\,dt = \frac{\Gamma(p)\Gamma(p-1)}{\Gamma(2p-1)}$$

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

$$\Gamma(-x) = \frac{\Gamma(1-x)}{-x}$$

$$\Gamma(p-1) = \Gamma(-(1-p)) = \frac{\Gamma(x-1+p)}{p-1} = \frac{\Gamma(p)}{p-1}$$

(j) $$\int_0^\infty \frac{x^{p-1}}{1+x^a}\,dx = \frac{\pi}{a \sin(p\pi/a)}$$

$$\beta(x,y) = \int_0^\infty \frac{t^{x-1}}{(1+t)^{x+y}}\,dt$$

$$x = (x^a)^{\frac{1}{a}}$$

$$\int_0^\infty \frac{(x^a)^{\frac{1}{a}\cdot p-1}}{1+x^a}\,dx$$

Let $u = x^a$

$$du = a x^{a-1}\,dx$$

$$u^{\frac{a-1}{a}} = x^{a-1}$$

$$\int_0^\infty \frac{(u)^{\frac{p-1}{a}}}{(1+u)^1}\,\frac{du}{a\,u^{\frac{a-1}{a}}}$$

$$\int_0^\infty \frac{(u)^{\frac{p-1}{a}}}{(1+u)^{1}} \frac{du}{a\,u^{\frac{a-1}{a}}}$$

$$\frac{p-1-a+1}{a} = \frac{p-a}{a}$$

$$\frac{1}{a}\int_0^\infty \frac{(u)^{\frac{p-a}{a}}}{(1+u)^{1}}\,du$$

$$\beta(x,y) = \int_b^\infty \frac{u^{x-1}}{(1+u)^{x+y}}\,dy$$

$$x-1 = \frac{p}{a} - 1$$

$$x = \frac{p}{a}$$

$$\frac{1}{a}\beta\left(\frac{p}{a}, 1-\frac{p}{a}\right)$$

$$1 = \frac{p}{a} + y \Rightarrow y = 1-\frac{p}{a}$$

$$= \frac{1}{a}\,\frac{\Gamma\left(\frac{p}{a}\right)\Gamma\left(1-\frac{p}{a}\right)}{\Gamma(1)}$$

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)}$$

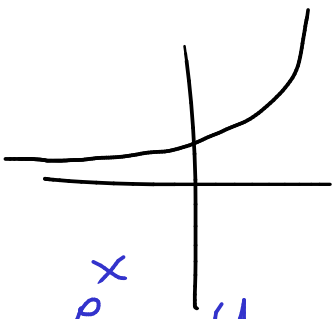
$$= \frac{1}{a}\,\frac{\pi}{\sin\left(\frac{\pi p}{a}\right)}$$

$$\beta(x,y) = \int_0^\infty \frac{t^{x-1}}{(1+t)^{x+y}}\,dt$$

$$\int_{-\infty}^{\infty} \frac{e^{2x}}{ae^{3x}+b}\,dx = \frac{2\pi}{3\sqrt{3}}\,a^{-2/3}b^{-1/3}$$

$$= \int_{-\infty}^{\infty} \frac{(e^x)}{a(e^x)^3+b}\,dx$$

$$= a\int_{-\infty}^{\infty} \frac{(e^x)^2}{(e^x)^3+\frac{b}{a}}\,dx$$



Let $e^x = u$

$u\,dx = du$

$$= a\int_0^\infty \frac{u^2}{u^3+b/a}\cdot\frac{du}{u}$$

$$= a\int_b^\infty \frac{u}{u^3+\frac{b}{a}}\,du$$

$$= \frac{a^2}{b}\,\frac{b}{a}\int_0^\infty \frac{\frac{a}{b}(u^3)^{\frac{1}{3}}}{\frac{a}{b}u^3+1}\,du$$

Let $\frac{a}{b}u^3 = t \Rightarrow \frac{3a}{b}u^2\,du = dt$

$$u^2 = \left(\frac{bt}{a}\right)^{\frac{2}{3}}$$

$$du = \frac{b\,dt}{3a\left(\frac{bt}{a}\right)^{2/3}}$$

$$\frac{1}{2}\beta(x,y) = \int_0^{\pi/2} \sin^{2x-1}\varphi\,\cos^{2y-1}\varphi\,d\varphi$$

$$\int_0^{\pi/2} \sqrt{\sin 2x}\,dx \quad \rightarrow \quad \int_0^{\pi/2} \sin^{\frac{1}{2}}(2x)\,dx$$

Let $\boxed{\varphi = 2x} \Rightarrow d\varphi = 2\,dx$

$$\int_0^{\pi} \sin^{\frac{1}{2}}(\varphi)\,\frac{d\varphi}{2} = \frac{1}{2}\int_0^{\pi} \sin^{\frac{1}{2}}(\varphi)\,d\varphi$$

$$= \frac{2}{2}\int_0^{\pi/2} \sin^{\frac{1}{2}}(\varphi)\,d\varphi$$

$$\int_0^{\pi/2} \sin^{\frac{1}{2}}\varphi \, d\varphi \cos^{0}\varphi$$

$$\beta(x,y) = 2\int_0^{\frac{\pi}{2}} \sin^{2x-1}\varphi \cos^{2y-1}\varphi \, d\varphi$$

$$\frac{1}{2}\beta\left(\frac{3}{4}, \frac{1}{2}\right)$$

$$= \frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{3}{4}+\frac{1}{2}\right)}$$

$$2x-1 = \frac{1}{2}$$

$$x = \frac{3}{4}$$

$$2y-1 = 0$$

$$y = \frac{1}{2}$$